

Cisco IOS & Networking Mastery Notes

From Zero to Advanced — CLI Configuration, Routing, Switching & Security

*A complete, hands-on reference for mastering Cisco IOS
and computer networking fundamentals through advanced topics*

Compiled Study Notes

September 1, 2026

Covers: OSI/TCP-IP fundamentals, subnetting, IOS CLI, switching (VLANs, STP, EtherChannel), routing (Static, RIP, EIGRP, OSPF, BGP), ACLs, NAT, DHCP, FHRP, Layer 2 & device security, WAN/VPN technologies, QoS, IPv6, file/image management, troubleshooting, and automation.

Contents

1	Networking Foundations	5
1.1	The OSI Model (7 Layers)	5
1.2	TCP/IP Model (4/5 Layers)	5
1.3	Encapsulation / De-encapsulation	5
1.4	TCP vs UDP	6
1.5	IPv4 Addressing	6
1.6	Subnetting (CIDR)	6
1.7	Switching Concepts	6
1.8	Routing Concepts	7
1.8.1	Administrative Distance (AD)	7
2	Cisco IOS Fundamentals	8
2.1	What is IOS?	8
2.2	Accessing the Device	8
2.3	CLI Modes (Hierarchy)	8
2.4	Getting Help & Navigation	8
2.5	Command Abbreviation & Do-command	9
2.6	Configuration Files	9
2.7	show and clear — the Backbone Commands	9
2.8	Basic Global Settings	10
3	Initial Device Configuration (Full Walkthrough)	11
3.1	SSH Setup (Detailed)	11
3.2	Local User Database & Privilege Levels	11
3.3	Banners	11
3.4	Saving, Backing Up, and Restoring Configs	11
3.5	Clock, NTP, and Time Zone	12
3.6	CDP & LLDP (Neighbor Discovery)	12
4	Interfaces & IP Addressing	13
4.1	Interface Naming	13
4.2	Basic Interface Configuration	13
4.3	Loopback Interfaces	13
4.4	IPv4 Address Assignment Methods	13
4.5	Secondary IP Addresses	13
4.6	Interface Troubleshooting States	14
5	Switching: VLANs, Trunking, STP, EtherChannel	15
5.1	VLANs	15
5.2	Trunking (802.1Q)	15
5.3	VTP (VLAN Trunking Protocol)	15
5.4	Spanning Tree Protocol (STP)	16
5.5	EtherChannel (Link Aggregation)	16

6	Inter-VLAN Routing	18
6.1	Router-on-a-Stick	18
6.2	SVI (Switched Virtual Interface) on a Layer 3 Switch	18
6.3	Routed Ports on L3 Switches	18
7	Static & Default Routing	19
7.1	Static Route Syntax	19
7.2	Verifying	19
8	Dynamic Routing Protocols	20
8.1	RIP (Routing Information Protocol) — legacy	20
8.2	EIGRP (Enhanced Interior Gateway Routing Protocol)	20
8.3	OSPF (Open Shortest Path First)	21
8.4	BGP (Border Gateway Protocol)	22
8.5	Route Redistribution (mixing protocols)	22
9	Access Control Lists (ACLs)	23
9.1	Standard ACLs	23
9.2	Extended ACLs	23
9.3	Named ACLs	23
9.4	Key Rules & Best Practices	23
9.5	Time-Based ACLs	24
9.6	Reflexive & Dynamic ACLs (advanced)	24
10	NAT / PAT	25
10.1	Static NAT (1:1 mapping)	25
10.2	Dynamic NAT (pool-based, many-to-many, no port translation)	25
10.3	PAT / NAT Overload (many-to-one, most common)	25
10.4	Verifying & Troubleshooting NAT	25
10.5	Port Forwarding (Static NAT with port)	25
11	DHCP	26
11.1	DHCP Server on IOS	26
11.2	DHCP Relay (IP Helper)	26
11.3	DORA Process	26
12	First Hop Redundancy Protocols (FHRP)	27
12.1	HSRP (Hot Standby Router Protocol) — Cisco proprietary	27
12.2	VRRP (Virtual Router Redundancy Protocol) — open standard	27
12.3	GLBP (Gateway Load Balancing Protocol) — Cisco proprietary	27
13	Layer 2 Security	28
13.1	Port Security	28
13.2	DHCP Snooping	28
13.3	Dynamic ARP Inspection (DAI)	28
13.4	IP Source Guard	29
13.5	Storm Control	29
13.6	VLAN Hopping Mitigations	29
14	Device Security & AAA	30
14.1	Password Best Practices	30
14.2	AAA (Authentication, Authorization, Accounting)	30
14.3	802.1X Port-Based Authentication	30
14.4	Control Plane Policing (CoPP) — advanced	30

15 WAN Technologies	32
15.1 PPP (Point-to-Point Protocol)	32
15.2 HDLC	32
15.3 Frame Relay (legacy WAN, still tested/asked about)	32
15.4 GRE Tunnels	32
15.5 Site-to-Site IPsec VPN (Crypto Map method, classic)	32
15.6 DMVPN (advanced overlay, brief overview)	33
16 QoS Fundamentals	34
16.1 Why QoS	34
16.2 Classification & Marking	34
16.3 Queuing (output congestion management)	34
16.4 Policing vs Shaping	34
16.5 Trust Boundaries	35
17 IPv6	36
17.1 Address Format	36
17.2 Address Types	36
17.3 Enabling & Configuring IPv6	36
17.4 Static & Dynamic IPv6 Routing	36
17.5 SLAAC & DHCPv6	36
18 IOS File & Image Management	38
18.1 File Systems	38
18.2 Backing Up / Upgrading IOS Image	38
18.3 Boot System Order & Config Register	38
18.4 Cisco IOS File Naming Convention (example)	38
19 Password Recovery & Licensing	39
19.1 Password Recovery (Generic Procedure)	39
19.2 Licensing (modern IOS/IOS-XE — Smart Licensing)	39
20 Monitoring, Logging & Troubleshooting	40
20.1 Logging	40
20.2 SNMP	40
20.3 Diagnostic Commands	40
20.4 The "OSI Layered" Troubleshooting Method	41
20.5 IP SLA (advanced monitoring)	41
21 Advanced Routing: Redistribution, Route-Maps, VRF-Lite	42
21.1 Route-Maps	42
21.2 Policy-Based Routing (PBR)	42
21.3 VRF-Lite (Virtual Routing and Forwarding)	42
21.4 MPLS & VXLAN (conceptual overview)	43
22 Automation: Netmiko, EEM, and Modern IOS-XE	44
22.1 Embedded Event Manager (EEM)	44
22.2 Netmiko (Python-based automation — industry standard)	44
22.3 RESTCONF / NETCONF (IOS-XE, model-driven programmability)	44
23 Appendix: Cheat Sheets	45
23.1 Subnetting Quick Reference	45
23.2 Common Well-Known Ports	45
23.3 Command Reference by Task	46

23.4 Suggested Mastery Path 46

Chapter 1

Networking Foundations

1.1 The OSI Model (7 Layers)

Layer	Name	PDU	Function	Example Protocols
7	Application	Data	User-facing services	HTTP, FTP, DNS, SMTP
6	Presentation	Data	Encryption, compression, formatting	SSL/TLS, JPEG
5	Session	Data	Session establishment/tear-down	NetBIOS, RPC
4	Transport	Segment	End-to-end delivery, reliability	TCP, UDP
3	Network	Packet	Logical addressing, routing	IP, ICMP, OSPF
2	Data Link	Frame	Physical addressing, framing	Ethernet, PPP, VLAN (802.1Q)
1	Physical	Bit	Electrical/optical transmission	Cabling, connectors, signaling

Mnemonic: All People Seem To Need Data Processing.

1.2 TCP/IP Model (4/5 Layers)

TCP/IP Layer	Maps to OSI
Application	5, 6, 7
Transport	4
Internet	3
Network Access (Link + Physical)	1, 2

1.3 Encapsulation / De-encapsulation

Data → Segment (L4, adds port number) → Packet (L3, adds IP) → Frame (L2, adds MAC) → Bits (L1). De-encapsulation is the reverse on the receiving device. Every device that forwards a frame at L2 (switch) only inspects the MAC address; every device that routes at L3 (router) strips and rebuilds the L2 frame each hop while the L3 packet's source/destination IP stays the same end-to-end (unless NAT is involved).

1.4 TCP vs UDP

- **TCP**: connection-oriented, 3-way handshake (SYN, SYN-ACK, ACK), sequencing, acknowledgment, flow control (windowing), retransmission, teardown (FIN/ACK $\times 4$ or RST). Used by HTTP, FTP, SSH, Telnet, SMTP.
- **UDP**: connectionless, no handshake, no guaranteed delivery, low overhead. Used by DNS, DHCP, TFTP, VoIP (RTP), SNMP, syslog.

1.5 IPv4 Addressing

32-bit address, 4 octets, dotted decimal notation (e.g., 192.168.1.1).

Class	Range	Default Prefix
A	0.0.0.0 – 127.255.255.255	/8
B	128.0.0.0 – 191.255.255.255	/16
C	192.0.0.0 – 223.255.255.255	/24
D	224.0.0.0 – 239.255.255.255	Multicast
E	240.0.0.0 – 255.255.255.255	Experimental

- **Private ranges (RFC 1918)**: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- **Loopback**: 127.0.0.0/8. **APIPA**: 169.254.0.0/16.

1.6 Subnetting (CIDR)

- CIDR notation: /prefix length = number of network bits.
- Subnet mask examples: /24 = 255.255.255.0, /25 = .128, /26 = .192, /27 = .224, /28 = .240, /29 = .248, /30 = .252, /32 = 255.255.255.255 (host route).
- **Magic number method**: $256 - (\text{last non-255 octet of mask}) = \text{block size}$.
Example: /27 $\rightarrow$ mask 255.255.255.224 $\rightarrow$ block size = $256 - 224 = 32 \rightarrow$ subnets: .0, .32, .64, .96, .128, .160, .192, .224.
- Usable hosts per subnet = $2^{(32-\text{prefix})} - 2$ (subtract network & broadcast). /30 = 2 usable hosts, ideal for point-to-point WAN links.
- **VLSM** (Variable Length Subnet Masking): subnetting a subnet — assign the largest host requirement first, smallest last, to conserve address space.
- **Route summarization** (supernetting): combine multiple contiguous subnets into one larger CIDR block advertised upstream, reducing routing table size.

1.7 Switching Concepts

- Switches operate at Layer 2, forwarding frames using the **MAC address table (CAM table)** learned dynamically from source MAC + ingress port.
- Unknown unicast, broadcast, and multicast frames are **flooded** out all ports except the ingress port (within the same VLAN/broadcast domain).

- **Collision domain:** segment where frames can collide (each switch port is its own collision domain in full duplex, eliminating collisions).
- **Broadcast domain:** extent to which broadcasts propagate; bounded by routers or VLAN boundaries.

1.8 Routing Concepts

Routers operate at Layer 3, forwarding packets based on the **routing table**, using the longest prefix match.

1.8.1 Administrative Distance (AD)

Trustworthiness of a route source (lower = more preferred):

Source	AD
Connected	0
Static	1
EIGRP summary	5
External BGP	20
Internal EIGRP	90
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200
Unknown / Unreachable	255

Metric: cost value used to select the best path within the *same* protocol (hop count for RIP, bandwidth/delay composite for EIGRP, cost = 100 Mbps / BW for OSPF).

Chapter 2

Cisco IOS Fundamentals

2.1 What is IOS?

Cisco **Internetwork Operating System (IOS)** is the CLI-driven OS running on Cisco routers/switches. Newer platforms may run **IOS-XE** (Linux-based, modular, supports containers) but the classic CLI syntax learned here applies almost identically.

2.2 Accessing the Device

- **Console port**: direct physical/serial (RJ45/USB) connection, required for initial setup, out-of-band, works even with no network config (default 9600 8N1 baud).
- **Telnet**: remote, unencrypted (avoid in production).
- **SSH**: remote, encrypted — preferred.
- **AUX port**: legacy modem access.

2.3 CLI Modes (Hierarchy)

User EXEC	Router>	limited, show-only, no config
Privileged EXEC	Router#	full show/debug/clear, "enable"
Global Config	Router(config)#	"configure terminal"
Interface Config	Router(config-if)#	"interface gi0/0"
Line Config	Router(config-line)#	"line vty 0 4" / "line console 0"
Router Config	Router(config-router)#	"router ospf 1"
VLAN Config	Router(config-vlan)#	"vlan 10"
Sub-interface	Router(config-subif)#	"interface gi0/0.10"

Move up one level: `exit`. Jump straight to privileged exec from anywhere: `end` or `Ctrl+Z`.

2.4 Getting Help & Navigation

Command / Key	Function
?	Context-sensitive help (list commands or arguments)
Tab	Auto-complete a partial command
Ctrl+A	Move cursor to beginning of line
Ctrl+E	Move cursor to end of line
Ctrl+R	Redisplay/refresh current line
Ctrl+W	Erase a word
Ctrl+U	Erase entire line
Ctrl+Shift+6, then X	Break/abort an ongoing process (e.g. stuck ping/-traceroute)

Command / Key	Function
Ctrl+Z	Exit to privileged EXEC from any config mode
Up/Down arrow, Ctrl+P/Ctrl+N	Cycle through command history
show history	List command history buffer
terminal history size <n>	Increase history buffer (default 10)

2.5 Command Abbreviation & Do-command

IOS allows shortest-unambiguous abbreviation: `conf t` = `configure terminal`, `sh run` = `show running-config`, `int gi0/0` = `interface gigabitEthernet0/0`.

From any config sub-mode, prefix `do` to run an EXEC-level command without leaving config mode:

```
Router(config-if)#do show ip interface brief
```

2.6 Configuration Files

- **running-config**: active configuration in RAM; lost on reload unless saved.
- **startup-config**: saved configuration in NVRAM; loaded on boot.

```
show running-config          ! (or: show run)
show startup-config          ! (or: show start)
copy running-config startup-config ! (or: write memory / wr)
copy startup-config running-config
write erase                  ! erases startup-config
erase startup-config
reload                      ! reboots device
```

2.7 show and clear — the Backbone Commands

```
show version                ! IOS version, uptime, hardware, config register
show running-config
show startup-config
show interfaces [int]       ! detailed L1/L2/L3 interface stats
show ip interface brief     ! quick status/IP summary
show ip route               ! routing table
show mac address-table      ! CAM table (switches)
show vlan brief
show cdp neighbors [detail]
show lldp neighbors [detail]
show inventory
show processes cpu
show memory statistics
show flash:
show logging
show arp
show users
show sessions
```

2.8 Basic Global Settings

```
enable
configure terminal
hostname R1
no ip domain-lookup          ! stop CLI from resolving mistyped commands via DNS
banner motd #Unauthorized access is prohibited#
banner login #Authorized users only#
enable secret Cisco123!      ! MD5/SHA-256 hashed enable password (preferred)
service password-encryption  ! weakly (type 7) encrypts plaintext passwords
line console 0
  password ConsolePass
  login
  exec-timeout 10 0          ! 10 min timeout, avoid idle console sessions
  logging synchronous       ! prevents log messages interrupting typed commands
exit
line vty 0 4
  password VtyPass
  login
  transport input ssh        ! restrict remote access to SSH only
exit
end
copy run start
```

Chapter 3

Initial Device Configuration (Full Walk-through)

3.1 SSH Setup (Detailed)

```
hostname R1
ip domain-name lab.local          ! required before generating RSA keys
crypto key generate rsa modulus 2048 ! generates the key pair, enables SSH
ip ssh version 2
username admin privilege 15 secret Str0ngP0ss
line vty 0 4
  transport input ssh
  login local                    ! authenticate against local username DB
exit
```

Verify: `show ip ssh`, `show crypto key mypubkey rsa`.

3.2 Local User Database & Privilege Levels

```
username netadmin privilege 15 secret Cisco@123
privilege exec level 5 show running-config ! custom privilege mapping (advanced)
```

Privilege levels: 0 (minimal: logout, enable, disable, help), 1 (user exec default), 15 (full privileged exec). Levels 2–14 are custom and rarely used without **privilege** command mappings.

3.3 Banners

```
banner motd ^C
*****
Authorized Access Only
*****
^C
```

Delimiting character (e.g. `^C` or `#`) must not appear inside the banner text itself.

3.4 Saving, Backing Up, and Restoring Configs

```
copy running-config startup-config
copy running-config tftp:
copy tftp: running-config
copy running-config flash:backup-config
copy startup-config scp:
```

Best practice: always back up config before major changes; use a TFTP/SCP server or a version-controlled text export.

3.5 Clock, NTP, and Time Zone

```
clock set 14:30:00 1 September 2026
clock timezone IST 5 30
ntp server 192.168.1.1
ntp master 3                ! act as an NTP source (usually only on a hub device)
show clock
show ntp status
show ntp associations
```

3.6 CDP & LLDP (Neighbor Discovery)

```
cdp run                    ! global enable (default on)
no cdp run                 ! disable globally for security on edge-facing devices
interface gi0/1
  no cdp enable            ! disable per interface
lldp run
show cdp neighbors detail
show lldp neighbors detail
```

CDP is Cisco-proprietary L2 discovery; LLDP is the vendor-neutral IEEE 802.1AB standard — use LLDP in mixed-vendor environments.

Chapter 4

Interfaces & IP Addressing

4.1 Interface Naming

Examples: FastEthernet0/1, GigabitEthernet0/0, TenGigabitEthernet1/0/1, Serial0/0/0, Loopback0, Vlan10 (SVI), Port-channel1, Tunnel0.

4.2 Basic Interface Configuration

```
interface gigabitEthernet0/0
  description Link-to-Core-SW
  ip address 192.168.10.1 255.255.255.0
  no shutdown          ! interfaces are admin DOWN by default on routers
  duplex full
  speed 1000
exit
```

Verify:

```
show ip interface brief
show interfaces gi0/0
show interfaces status      ! switches - shows connected/notconnect/err-disabled
show controllers serial 0/0/0 ! shows DCE/DTE, needed for clock rate
```

On a serial link, the **DCE end** must set clocking: `clock rate 64000` (bps) inside interface config.

4.3 Loopback Interfaces

```
interface loopback0
  ip address 10.10.10.1 255.255.255.255
```

Always up as long as the device is up (no physical layer) — used for router-IDs (OSPF/BGP), management, and testing.

4.4 IPv4 Address Assignment Methods

```
ip address 192.168.1.1 255.255.255.0      ! static
ip address dhcp                          ! interface obtains via DHCP client
ip address negotiated                     ! PPP/PPPoE negotiated address
```

4.5 Secondary IP Addresses

```
interface gi0/0
  ip address 192.168.1.1 255.255.255.0
  ip address 192.168.2.1 255.255.255.0 secondary
```

Used when multiple subnets must share one physical segment (legacy technique; prefer sub-interfaces/VLANs where possible).

4.6 Interface Troubleshooting States

show interfaces output	Meaning
up, line protocol up	Fully operational
administratively down, down	shutdown applied
down, down	Physical/Layer 1 problem (cable, far end down)
up, line protocol down	L2 issue: encapsulation mismatch, keepalives, no clock rate on serial
err-disabled	Switch disabled port due to violation (port-security, BPDU guard, storm-control) — fix cause then shutdown/no shutdown , or configure errdisable recovery

```
errdisable recovery cause psecure-violation
errdisable recovery interval 60
```

Chapter 5

Switching: VLANs, Trunking, STP, EtherChannel

5.1 VLANs

VLANs create separate broadcast domains on a shared physical switch fabric.

```
vlan 10
  name SALES
vlan 20
  name ENGINEERING
exit
interface fa0/1
  switchport mode access
  switchport access vlan 10
```

Verify: `show vlan brief`, `show interfaces fa0/1 switchport`.

Voice VLAN (for IP phones sharing a port with a PC):

```
interface fa0/2
  switchport mode access
  switchport access vlan 10
  switchport voice vlan 100
```

5.2 Trunking (802.1Q)

Trunks carry multiple VLANs between switches, tagging frames with a 4-byte 802.1Q header (VLAN ID 1–4094). The **native VLAN** (default VLAN 1) is sent untagged — must match on both ends or you get VLAN mismatch issues/security risk.

```
interface gi0/1
  switchport trunk encapsulation dot1q      ! needed on switches supporting ISL too
  switchport mode trunk
  switchport trunk native vlan 999          ! change from default VLAN1 (best practice)
  switchport trunk allowed vlan 10,20,30    ! prune unnecessary VLANs
  switchport trunk allowed vlan add 40
```

DTP (Dynamic Trunking Protocol): auto-negotiates trunk/access mode (dynamic auto/dynamic desirable). Best practice: disable DTP with `switchport nonegotiate` and hard-set modes explicitly to avoid VLAN-hopping attacks.

5.3 VTP (VLAN Trunking Protocol)

Propagates VLAN database across switches in a domain.

```
vtp domain CORP
vtp mode server      ! server (default) | client | transparent | off
```



```
vtp password Cisco123
vtp version 2
show vtp status
```

- **Server:** can create/modify/delete VLANs, propagates changes.
- **Client:** cannot modify VLANs locally, only receives.
- **Transparent:** forwards VTP ads but keeps its own independent local VLAN DB — recommended in modern networks to avoid the "VTP revision number disaster" (a switch with a higher revision number can wipe out the whole domain's VLANs). Mitigate by always resetting revision to 0 before connecting a new switch.

5.4 Spanning Tree Protocol (STP)

Prevents Layer 2 loops in redundant topologies by blocking redundant paths.

- **Root Bridge election:** lowest Bridge ID (Priority + MAC) wins. Default priority 32768.
- **Port Roles:** Root Port (best path to root, one per non-root switch), Designated Port (best path per segment), Blocking/Alternate Port (loop prevention).
- **Port States (classic 802.1D):** Blocking → Listening → Learning → Forwarding (each transition ~15–20s, total ~30–50s convergence — slow!).

```
spanning-tree mode pvst           ! PVST+ (default, per-VLAN)
spanning-tree mode rapid-pvst     ! RSTP per VLAN - fast convergence (sub-second)
spanning-tree mode mst            ! Multiple Spanning Tree - maps VLANs to instances
spanning-tree vlan 10 priority 4096 ! influence root election (lower=preferred, incr.
4096)
spanning-tree vlan 10 root primary ! auto-sets priority lower than current root
spanning-tree vlan 10 root secondary
```

STP Enhancements (must-know for real deployments):

```
interface fa0/1
  spanning-tree portfast           ! skips listening/learning (end-hosts only!)
  spanning-tree bpduguard enable   ! err-disables port if BPDU received on PortFast
  port
spanning-tree portfast bpduguard default ! global enable for all portfast ports
spanning-tree guard root           ! prevents downstream switch from becoming root
spanning-tree loopguard default
```

Verify: show spanning-tree, show spanning-tree vlan 10, show spanning-tree interface fa0/1 detail.

5.5 EtherChannel (Link Aggregation)

Bundles multiple physical links into one logical link for bandwidth + redundancy; STP treats the bundle as a single link (no blocking).

```
interface range gi0/1 - 2
  channel-group 1 mode active      ! LACP (IEEE standard) - active/passive
  channel-group 1 mode desirable  ! PAgP (Cisco proprietary) - desirable/auto
  channel-group 1 mode on         ! static, no negotiation (not recommended)
interface port-channel 1
```

```
switchport mode trunk
```

Rule: all bundled physical ports must match — speed, duplex, VLAN/trunk config, STP settings — or bundling fails.

Verify: `show etherchannel summary`, `show etherchannel port-channel`.

```
port-channel load-balance src-dst-ip
```

Chapter 6

Inter-VLAN Routing

6.1 Router-on-a-Stick

One physical router interface, sub-interfaced per VLAN, connected to a trunk port on the switch.

```
interface gi0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
interface gi0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
interface gi0/0.999
  encapsulation dot1Q 999 native      ! native VLAN sub-interface, no tag added
```

Switch side:

```
interface gi0/1
  switchport mode trunk
```

6.2 SVI (Switched Virtual Interface) on a Layer 3 Switch

Preferred in modern designs (faster, no single trunk bottleneck):

```
ip routing                                ! enable L3 routing capability on the switch
interface vlan 10
  ip address 192.168.10.1 255.255.255.0
  no shutdown
interface vlan 20
  ip address 192.168.20.1 255.255.255.0
```

6.3 Routed Ports on L3 Switches

```
interface gi0/1
  no switchport                          ! converts a switch port into a pure L3 port
  ip address 10.0.0.1 255.255.255.252
```

Chapter 7

Static & Default Routing

7.1 Static Route Syntax

```
ip route <destination-network> <subnet-mask> <next-hop-ip | exit-interface> [AD] [name X]
ip route 192.168.20.0 255.255.255.0 10.0.0.2
ip route 192.168.20.0 255.255.255.0 gi0/0/1           ! exit-interface (P2P links only)
ip route 0.0.0.0 0.0.0.0 203.0.113.1                 ! default route (gateway of last
resort)
ip route 192.168.20.0 255.255.255.0 10.0.0.2 200      ! floating static (higher AD =
backup)
```

- Prefer **next-hop IP** style on multi-access (Ethernet) segments to avoid proxy-ARP/recursive lookup issues; exit-interface style is fine (even preferred) only on true point-to-point (serial/PPP) links.
- **Floating static routes** act as backup: configured with a higher AD than the primary dynamic/static route so they're only installed in the RIB when the primary fails.

7.2 Verifying

```
show ip route
show ip route static
show ip route 192.168.20.0
traceroute 192.168.20.5
```

Route table codes: C=connected, S=static, O=OSPF, D=EIGRP, R=RIP, B=BGP, L=local (host route /32 auto-created per interface), *=candidate default, >=next hop marker.

Chapter 8

Dynamic Routing Protocols

8.1 RIP (Routing Information Protocol) — legacy

Distance-vector, hop-count metric (max 15, 16 = unreachable), updates every 30s, slow convergence.

```
router rip
  version 2
  network 192.168.1.0
  no auto-summary
  passive-interface gi0/0           ! suppress RIP updates out user-facing interfaces
```

8.2 EIGRP (Enhanced Interior Gateway Routing Protocol)

Cisco advanced distance-vector (hybrid). Uses **DUAL (Diffusing Update Algorithm)** for loop-free, fast convergence. Metric = composite of bandwidth + delay (optionally load/reliability, rarely used).

```
router eigrp 100                     ! classic mode, AS number must match neighbors
  network 192.168.1.0 0.0.0.255
  no auto-summary
  eigrp router-id 1.1.1.1
  passive-interface gi0/0
```

Named mode (modern, recommended):

```
router eigrp NAMED-PROC
  address-family ipv4 unicast autonomous-system 100
  network 192.168.1.0 0.0.0.255
  af-interface gi0/0
  passive-interface
  exit-af-interface
  exit-address-family
```

Key terms: **Successor** (best route, installed in RIB), **Feasible Successor** (backup route, pre-computed, satisfies feasibility condition — instant failover with no recomputation), **Feasibility Condition** (Reported Distance of neighbor < Feasible Distance of current successor).

Verify:

```
show ip eigrp neighbors
show ip eigrp topology
show ip route eigrp
show ip protocols
```

Authentication: `key chain`, then under interface: `ip authentication mode eigrp 100 md5 / ip authentication key-chain eigrp 100 CHAIN1.`

8.3 OSPF (Open Shortest Path First)

Link-state, industry standard. Uses **Dijkstra's SPF algorithm**. Metric = cost = reference-bandwidth / interface-bandwidth (default reference 100 Mbps — adjust with **auto-cost reference-bandwidth** on modern high-speed links).

```
router ospf 1
  router-id 1.1.1.1
  network 192.168.1.0 0.0.0.255 area 0
  network 10.0.0.0 0.0.0.3 area 0
  passive-interface gi0/0
  auto-cost reference-bandwidth 10000      ! in Mbps
```

Per-interface alternative (modern, especially on L3 switches):

```
interface gi0/0
  ip ospf 1 area 0
  ip ospf cost 20
  ip ospf priority 255                ! influences DR/BDR election, higher wins
  ip ospf hello-interval 5
  ip ospf dead-interval 20
```

OSPF Concepts

- **Areas:** Area 0 (backbone) is mandatory in multi-area designs; all other areas must connect to Area 0 (directly or via virtual-link).
- **DR/BDR election:** on multi-access (broadcast/NBMA) segments only, to reduce adjacency flooding; elected by highest priority, tie-break highest Router ID. Point-to-point serial links skip DR/BDR.
- **LSA Types:** Type 1 (Router), Type 2 (Network — from DR), Type 3 (Summary/inter-area, from ABR), Type 4 (ASBR Summary), Type 5 (External, from ASBR), Type 7 (NSSA External, converted to Type 5 at the NSSA ABR).
- **Area types:** Standard, Stub (blocks Type 5, injects default route), Totally Stubby (Cisco-proprietary, blocks Type 3+4+5), NSSA (allows limited external redistribution via Type 7).

```
area 10 stub
area 10 stub no-summary          ! totally stubby (on ABR only)
area 20 nssa
```

Router types: Internal (all interfaces in one area), Backbone (interface in area 0), ABR — Area Border Router (interfaces in multiple areas), ASBR — Autonomous System Boundary Router (redistributes external routes into OSPF).

Verify:

```
show ip ospf neighbor
show ip ospf interface brief
show ip ospf database
show ip route ospf
show ip protocols
```

OSPF neighbor states: Down → Attempt → Init → 2-Way → ExStart → Exchange → Loading → **Full** (fully synchronized).

8.4 BGP (Border Gateway Protocol)

Path-vector, Internet-scale routing. Used between Autonomous Systems (eBGP) or within large enterprises (iBGP). Not metric-based like IGPs — uses a **best-path selection algorithm** based on attributes.

```
router bgp 65001
  bgp router-id 1.1.1.1
  neighbor 203.0.113.2 remote-as 65002      ! eBGP - different AS
  neighbor 10.0.0.2 remote-as 65001        ! iBGP - same AS, needs full-mesh/RR
  network 192.168.1.0 mask 255.255.255.0
  neighbor 203.0.113.2 update-source loopback0
  neighbor 203.0.113.2 password Cisco123
```

BGP Best-Path Selection Order (simplified)

1. Highest Weight (Cisco-local only)
2. Highest Local Preference
3. Locally originated route preferred
4. Shortest AS-Path
5. Lowest Origin type (IGP < EGP < Incomplete)
6. Lowest MED (Multi-Exit Discriminator)
7. eBGP over iBGP
8. Lowest IGP metric to next hop
9. Oldest route (most stable)
10. Lowest Router ID
11. Lowest neighbor IP

Verify: `show ip bgp summary`, `show ip bgp`, `show ip bgp neighbors`.

8.5 Route Redistribution (mixing protocols)

```
router ospf 1
  redistribute eigrp 100 subnets metric 50 metric-type 1
router eigrp 100
  redistribute ospf 1 metric 10000 100 255 1 1500
```

Always watch for **routing loops** when redistributing bidirectionally between two protocols at multiple points — use route-maps + tagging to prevent re-injection.

Chapter 9

Access Control Lists (ACLs)

9.1 Standard ACLs

Filter by source IP only, numbers 1–99, 1300–1999.

```
access-list 10 permit 192.168.1.0 0.0.0.255
access-list 10 deny any
interface gi0/1
 ip access-group 10 out
```

Rule: place standard ACLs **as close to the destination as possible** (since they can't filter by destination, applying near source could block unintended traffic). Wildcard mask = inverse of subnet mask (0 = must match, 1 = don't care). 0.0.0.255 matches the first 3 octets exactly.

9.2 Extended ACLs

Filter by source, destination, protocol, port — numbers 100–199, 2000–2699.

```
access-list 110 permit tcp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 eq 80
access-list 110 permit tcp any any established      ! allow return traffic
access-list 110 deny ip any any log                 ! implicit deny is always last
interface gi0/0
 ip access-group 110 in
```

Rule: place extended ACLs **as close to the source as possible** to drop unwanted traffic early.

9.3 Named ACLs

Easier to edit — supports line numbers, insert/delete.

```
ip access-list extended BLOCK-TELNET
 10 deny tcp any any eq 23
 20 permit ip any any
```

Editing: no 10 removes just line 10, then add a new one with a chosen sequence number, e.g. 15 deny tcp host 10.1.1.1 any eq 23.

9.4 Key Rules & Best Practices

- ACLs are processed **top-down, first match wins** — order matters.
- Every ACL has an **implicit deny any** at the end (invisible but real).
- One ACL per interface, per direction, per protocol.
- Router-generated traffic is not filtered by an *inbound* ACL on that router unless via other mechanisms (e.g. Control Plane Policing).
- Verify: `show access-lists`, `show ip interface gi0/0` (shows applied ACLs).

9.5 Time-Based ACLs

```
time-range WORK-HOURS  
  periodic weekdays 9:00 to 18:00  
access-list 110 permit tcp any any eq 80 time-range WORK-HOURS
```

9.6 Reflexive & Dynamic ACLs (advanced)

- **Reflexive ACL:** auto-creates temporary return-traffic entries for sessions initiated from inside — stateful-like filtering without a full firewall.
- **Dynamic ("lock-and-key") ACL:** requires Telnet/SSH authentication before temporarily opening additional access.

Chapter 10

NAT / PAT

10.1 Static NAT (1:1 mapping)

```
ip nat inside source static 192.168.1.10 203.0.113.10
interface gi0/0
 ip nat inside
interface gi0/1
 ip nat outside
```

10.2 Dynamic NAT (pool-based, many-to-many, no port translation)

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat pool NATPOOL 203.0.113.10 203.0.113.20 netmask 255.255.255.0
ip nat inside source list 1 pool NATPOOL
```

10.3 PAT / NAT Overload (many-to-one, most common)

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface gi0/1 overload
```

10.4 Verifying & Troubleshooting NAT

```
show ip nat translations
show ip nat statistics
clear ip nat translation *
debug ip nat
```

Common issue: forgetting to mark both `ip nat inside` and `ip nat outside` on the correct interfaces — NAT silently does nothing without both.

10.5 Port Forwarding (Static NAT with port)

```
ip nat inside source static tcp 192.168.1.100 80 203.0.113.10 8080
```

Chapter 11

DHCP

11.1 DHCP Server on IOS

```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp pool LAN-POOL
  network 192.168.1.0 255.255.255.0
  default-router 192.168.1.1
  dns-server 8.8.8.8 8.8.4.4
  domain-name lab.local
  lease 7                                ! days (default) [hours] [minutes] | lease infinite
```

Verify: `show ip dhcp binding`, `show ip dhcp pool`, `show ip dhcp conflict`.

11.2 DHCP Relay (IP Helper)

For clients on a different subnet than the DHCP server:

```
interface vlan 20
  ip helper-address 192.168.1.5
```

`ip helper-address` also forwards other broadcast UDP services (TFTP, DNS, NetBIOS, etc.) by default unless restricted with `ip forward-protocol`.

11.3 DORA Process

Discover (client broadcast) → **O**ffer (server) → **R**quest (client broadcasts chosen offer) → **A**cknowledge (server confirms lease).

Chapter 12

First Hop Redundancy Protocols (FHRP)

Provide a resilient default gateway for hosts when the primary router fails.

12.1 HSRP (Hot Standby Router Protocol) — Cisco proprietary

```
interface gi0/0
  ip address 192.168.1.2 255.255.255.0
  standby 1 ip 192.168.1.1           ! virtual IP, configured on all group members
  standby 1 priority 110             ! higher wins active role (default 100)
  standby 1 preempt                  ! allow higher-priority router to reclaim Active
  standby 1 track gi0/1 decrement 20 ! object tracking - lower priority if uplink
  fails
  standby version 2
```

Virtual MAC: 0000.0C9F.Fxxx (v1) — one Active, one Standby, others Listen.

12.2 VRRP (Virtual Router Redundancy Protocol) — open standard

```
interface gi0/0
  vrrp 1 ip 192.168.1.1
  vrrp 1 priority 110
  vrrp 1 preempt
```

Terms: Master (active) / Backup (vs HSRP's Active/Standby). Virtual MAC: 0000.5E00.01xx.

12.3 GLBP (Gateway Load Balancing Protocol) — Cisco proprietary

All members can actively forward simultaneously (unlike HSRP/VRRP's single active router) via a single virtual IP but multiple virtual MACs (AVG assigns MACs to AVFs).

```
interface gi0/0
  glbp 1 ip 192.168.1.1
  glbp 1 load-balancing round-robin
```

Verify: `show standby brief`, `show vrrp brief`, `show glbp brief`.

Chapter 13

Layer 2 Security

13.1 Port Security

Limits/controls which MAC addresses can use a switch port.

```
interface fa0/1
  switchport mode access
  switchport port-security
  switchport port-security maximum 2
  switchport port-security mac-address sticky      ! learn dynamically & save
  switchport port-security mac-address 0011.2233.4455 ! static entry
  switchport port-security violation shutdown      ! shutdown | restrict | protect
```

- **shutdown**: err-disables port, sends SNMP/syslog trap (default, most common).
- **restrict**: drops offending traffic, increments violation counter, logs.
- **protect**: drops offending traffic silently, no logging.

Verify: `show port-security`, `show port-security interface fa0/1`, `show port-security address`.

13.2 DHCP Snooping

Prevents rogue DHCP servers by trusting only designated uplink ports.

```
ip dhcp snooping
ip dhcp snooping vlan 10,20
interface gi0/1
  ip dhcp snooping trust      ! uplink to legitimate DHCP server
interface fa0/1
  ip dhcp snooping limit rate 10 ! pps, mitigates DoS on untrusted ports
```

Builds the **DHCP snooping binding table** (IP-MAC-VLAN-Port-lease), which feeds Dynamic ARP Inspection and IP Source Guard.

13.3 Dynamic ARP Inspection (DAI)

Prevents ARP spoofing/man-in-the-middle by validating ARP packets against the DHCP snooping binding table.

```
ip arp inspection vlan 10,20
interface gi0/1
  ip arp inspection trust
```

13.4 IP Source Guard

Prevents IP spoofing by validating source IP/MAC against the DHCP snooping table, per port.

```
interface fa0/1
  ip verify source
```

13.5 Storm Control

Mitigates broadcast/multicast/unicast storms.

```
interface fa0/1
  storm-control broadcast level 50.00
  storm-control action shutdown
```

13.6 VLAN Hopping Mitigations

Disable DTP (`switchport nonegotiate`), set unused ports to a non-default "parking lot" VLAN and shut them down, never leave native VLAN as VLAN 1, disable auto-trunking, prune unused VLANs off trunks.

Chapter 14

Device Security & AAA

14.1 Password Best Practices

```
enable secret Cisco@123      ! preferred (hashed) over 'enable password'
service password-encryption  ! type 7 - weak, obscures from casual viewing
username admin secret Str0ng@1
```

Modern IOS supports stronger hashing: `enable algorithm-type scrypt secret <pw> / username admin algorithm-type scrypt secret <pw>`.

14.2 AAA (Authentication, Authorization, Accounting)

```
aaa new-model
aaa authentication login default local      ! fallback method list
aaa authentication login VTY-LOGIN group tacacs+ local
aaa authorization exec default group tacacs+ local
aaa accounting exec default start-stop group tacacs+
tacacs server TAC1
  address ipv4 10.1.1.5
  key Cisco123
line vty 0 4
  login authentication VTY-LOGIN
```

RADIUS alternative:

```
radius server RAD1
  address ipv4 10.1.1.6 auth-port 1812 acct-port 1813
  key Cisco123
aaa authentication login default group radius local
```

TACACS+ (TCP 49, encrypts entire packet, separates AAA) vs RADIUS (UDP 1812/1813, encrypts only password, combines authN+authZ) — TACACS+ preferred for device administration, RADIUS common for network access (802.1X, VPN).

14.3 802.1X Port-Based Authentication

```
aaa new-model
dot1x system-auth-control
interface fa0/1
  authentication port-control auto
  dot1x pae authenticator
```

14.4 Control Plane Policing (CoPP) — advanced

Protects the router's CPU from attack traffic.

```
class-map match-all CRITICAL
  match access-group 150
policy-map COPP-POLICY
  class CRITICAL
    police 8000 conform-action transmit exceed-action drop
control-plane
  service-policy input COPP-POLICY
```


Chapter 15

WAN Technologies

15.1 PPP (Point-to-Point Protocol)

```
interface serial0/0/0
  encapsulation ppp
  ppp authentication chap
  ppp chap hostname R1
  ppp chap password Cisco123
```

CHAP (Challenge Handshake, 3-way, password never sent in clear) preferred over PAP (2-way, plain-text). PPP supports **multilink (MLPPP)** to bundle multiple serial links: `interface multilink1`.

15.2 HDLC

Cisco's default serial encapsulation (`encapsulation hdlc`) — simpler, no authentication, only Cisco-to-Cisco compatible in its proprietary form.

15.3 Frame Relay (legacy WAN, still tested/asked about)

```
interface serial0/0/0
  encapsulation frame-relay
  frame-relay lmi-type ansi
interface serial0/0/0.100 point-to-point
  ip address 10.0.0.1 255.255.255.252
  frame-relay interface-dlci 100
```

Key terms: **DLCI** (Data Link Connection Identifier, locally significant), **LMI** (Local Management Interface — signaling), **PVC** (Permanent Virtual Circuit).

15.4 GRE Tunnels

Encapsulates a wide variety of protocols across an IP network — no encryption by itself.

```
interface tunnel0
  ip address 172.16.0.1 255.255.255.252
  tunnel source gi0/0
  tunnel destination 203.0.113.5
  tunnel mode gre ip          ! default mode, can be omitted
```

15.5 Site-to-Site IPsec VPN (Crypto Map method, classic)

```
crypto isakmp policy 10
  encryption aes 256
  hash sha256
  authentication pre-share
  group 14
```

```
crypto isakmp key Cisco123 address 203.0.113.5
crypto ipsec transform-set TSET esp-aes 256 esp-sha256-hmac
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
crypto map CMAP 10 ipsec-isakmp
    set peer 203.0.113.5
    set transform-set TSET
    match address 100
interface gi0/1
    crypto map CMAP
```

GRE-over-IPsec is common when you need routing protocols (OSPF/EIGRP) across the VPN, since plain IPsec (tunnel mode) can't carry multicast/broadcast routing hellos.

15.6 DMVPN (advanced overlay, brief overview)

Combines **mGRE** (multipoint GRE) + **NHRP** (Next Hop Resolution Protocol) + IPsec to build a dynamic hub-and-spoke or spoke-to-spoke VPN mesh without manually configuring every tunnel.

Chapter 16

QoS Fundamentals

16.1 Why QoS

Manages congestion, delay, jitter, and packet loss for latency-sensitive traffic (voice, video) by classifying, marking, queuing, and policing/shaping traffic.

16.2 Classification & Marking

```
class-map match-all VOICE
  match dscp ef
policy-map MARK-TRAFFIC
  class VOICE
    set dscp ef
interface gi0/0
  service-policy input MARK-TRAFFIC
```

Common markings: **DSCP EF** (Expedited Forwarding, voice), **AF** classes (Assured Forwarding, data), **CS** (Class Selector, legacy IP Precedence-compatible).

16.3 Queuing (output congestion management)

```
policy-map QOS-OUT
  class VOICE
    priority percent 20          ! LLQ - Low Latency Queuing, strict priority
  class DATA
    bandwidth percent 50
  class class-default
    fair-queue
interface gi0/0
  service-policy output QOS-OUT
```

16.4 Policing vs Shaping

- **Policing**: drops (or re-marks) traffic exceeding a rate immediately — no buffering, used for enforcing SLAs on ingress.
- **Shaping**: buffers/delays excess traffic to smooth bursts to a target rate — used on egress, especially toward slower WAN links.

```
policy-map POLICE-HTTP
  class HTTP
    police 1000000 conform-action transmit exceed-action drop
policy-map SHAPE-WAN
  class class-default
    shape average 512000
```

16.5 Trust Boundaries

Configure switches to trust or override markings from end-devices to prevent users from self-prioritizing traffic:

```
interface fa0/1
  mls qos trust device cisco-phone
  mls qos trust cos
```

Chapter 17

IPv6

17.1 Address Format

128-bit, 8 groups of 4 hex digits, :: compresses one run of consecutive zero groups (only once per address). Example: 2001:0db8:0000:0000:0000:0000:0001 → 2001:db8::1.

17.2 Address Types

- **Unicast Global (GUA):** 2000::/3 — routable on the Internet.
- **Link-Local (LLA):** fe80::/10 — auto-assigned, mandatory on every interface, never routed.
- **Unique Local (ULA):** fc00::/7 (typically fd00::/8) — private, like RFC 1918 for IPv6.
- **Multicast:** ff00::/8. No broadcast in IPv6 — replaced by multicast (e.g., ff02::1 = all-nodes, ff02::2 = all-routers).
- **Anycast:** same address assigned to multiple devices, nearest one responds.

17.3 Enabling & Configuring IPv6

```
ipv6 unicast-routing          ! required on routers to actually route IPv6
interface gi0/0
  ipv6 address 2001:db8:1::1/64
  ipv6 address fe80::1 link-local
  ipv6 enable                 ! generates link-local only, no GUA
  ipv6 address 2001:db8:1::/64 eui-64 ! auto-generates host portion from MAC
```

17.4 Static & Dynamic IPv6 Routing

```
ipv6 route 2001:db8:2::/64 2001:db8:1::2
ipv6 route ::/0 2001:db8:1::1
router ospfv3 1
  router-id 1.1.1.1
interface gi0/0
  ospfv3 1 area 0
router eigrp NAME
  address-family ipv6 unicast autonomous-system 100
  eigrp router-id 1.1.1.1
interface gi0/0
  ipv6 eigrp 100
```

17.5 SLAAC & DHCPv6

```
interface gi0/0
  ipv6 nd other-config-flag    ! tells hosts to also query DHCPv6 (stateless)
  ipv6 dhcp pool DHCPV6POOL
```

```
dns-server 2001:4860:4860::8888
domain-name lab.local
interface gi0/0
  ipv6 dhcp server DHCPV6POOL
```

Verify: `show ipv6 interface brief`, `show ipv6 route`, `show ipv6 neighbors` (equivalent of ARP, called **Neighbor Discovery Protocol / NDP**).

Chapter 18

IOS File & Image Management

18.1 File Systems

```
show file systems
dir flash:
dir nvram:
```

18.2 Backing Up / Upgrading IOS Image

```
copy flash: tftp:           ! backup current image
copy tftp: flash:           ! upload a new image
verify /md5 flash:new-image.bin ! confirm integrity
boot system flash:new-image.bin ! tell IOS which image to boot next
show boot
reload
```

18.3 Boot System Order & Config Register

```
config-register 0x2102      ! normal boot (default), reads startup-config
config-register 0x2142      ! ignores startup-config (used in password recovery)
show version                ! shows current config register value at the bottom
```

18.4 Cisco IOS File Naming Convention (example)

c2900-universalk9-mz.SPA.155-3.M4.bin → platform (c2900), feature set (universalk9), run-from-RAM/zip (mz), signed package (SPA), version (15.5(3)M4).

Chapter 19

Password Recovery & Licensing

19.1 Password Recovery (Generic Procedure)

1. Connect via console cable.
2. Power cycle the device; send a **break** signal during boot (within ~60 sec) to enter **ROMMON**.
3. `confreg 0x2142` (skip startup-config on boot) then `reset`/power-cycle.
4. Device boots with no startup-config → enter `enable` (no password).
5. `copy startup-config running-config` to restore the config into RAM.
6. Change passwords: `enable secret NewPass`, update line/vty passwords.
7. `config-register 0x2102` to restore normal boot behavior.
8. `copy running-config startup-config`, `reload`.

19.2 Licensing (modern IOS/IOS-XE — Smart Licensing)

```
show license summary
show license status
license smart register idtoken <token>
license boot level network-advantage add-on dna-advantage
```

Older platforms used `license install <file>` (right-to-use / permanent licenses via activation keys).

Chapter 20

Monitoring, Logging & Troubleshooting

20.1 Logging

```
logging buffered 16384
logging console
logging trap informational
logging host 192.168.1.100      ! syslog server
service timestamps log datetime msec
show logging
```

Syslog severity levels (0 = most severe): 0 Emergency, 1 Alert, 2 Critical, 3 Error, 4 Warning, 5 Notice, 6 Informational, 7 Debugging.

20.2 SNMP

```
snmp-server community public RO
snmp-server community private RW
snmp-server location DC1-Rack4
snmp-server contact netops@company.com
snmp-server host 192.168.1.100 version 2c public
snmp-server enable traps
```

Prefer **SNMPv3** for production (adds authentication + encryption):

```
snmp-server group ADMINGRP v3 priv
snmp-server user netadmin ADMINGRP v3 auth sha AuthPass123 priv aes 128 PrivPass123
```

20.3 Diagnostic Commands

```
ping 8.8.8.8
ping 8.8.8.8 source loopback0
! extended ping - interactively prompts for repeat count, size, source, etc.
traceroute 8.8.8.8
show ip route <destination>
show cdp neighbors detail
show interfaces gi0/0 | include (line protocol|packets)
show ip interface brief | exclude unassigned
debug ip icmp
debug ip routing
debug ip ospf adjacency
undebug all      ! (or "u all") - stop ALL debugging immediately
terminal monitor ! shows debug/log output on vty sessions
```

Debugging is CPU-intensive — never leave `debug all` running on a production device.

20.4 The "OSI Layered" Troubleshooting Method

1. **Physical:** cabling, `show interfaces` (up/up?), SFP/light status.
2. **Data Link:** VLAN assignment, trunk config, STP state, encapsulation match.
3. **Network:** IP addressing, subnet mask, routing table, ACLs blocking.
4. **Transport:** port blocked by firewall/ACL, NAT translation issue.
5. **Application:** DNS resolution, service actually running.

Work bottom-up (or top-down / divide-and-conquer) systematically rather than guessing.

20.5 IP SLA (advanced monitoring)

```
ip sla 1
  icmp-echo 8.8.8.8 source-interface gi0/0
  frequency 30
ip sla schedule 1 life forever start-time now
track 1 ip sla 1 reachability
ip route 0.0.0.0 0.0.0.0 203.0.113.1 track 1
```

Used to conditionally install/remove a route based on real reachability testing (common with FHRP + floating statics for automatic failover).

Chapter 21

Advanced Routing: Redistribution, Route-Maps, VRF-Lite

21.1 Route-Maps

The Swiss-army-knife of policy control — used for redistribution filtering, PBR (Policy-Based Routing), BGP attribute manipulation, NAT route filtering, etc.

```
ip prefix-list PL-10 seq 5 permit 192.168.10.0/24
route-map REDIST-FILTER permit 10
  match ip address prefix-list PL-10
  set metric 20
route-map REDIST-FILTER deny 20
!
router ospf 1
  redistribute eigrp 100 subnets route-map REDIST-FILTER
```

Route-maps are processed top-down like ACLs, first-match, implicit deny at the end (be careful — an empty/unmatched route-map denies everything by default).

21.2 Policy-Based Routing (PBR)

Route based on criteria other than destination IP.

```
access-list 101 permit ip 192.168.1.0 0.0.0.255 any
route-map PBR-MAP permit 10
  match ip address 101
  set ip next-hop 10.0.0.2
interface gi0/0
  ip policy route-map PBR-MAP
```

21.3 VRF-Lite (Virtual Routing and Forwarding)

Creates multiple independent routing tables on one device — used for multi-tenancy without MPLS.

```
ip vrf CUSTOMER_A
  rd 65000:1
interface gi0/0
  ip vrf forwarding CUSTOMER_A
  ip address 192.168.1.1 255.255.255.0
ip route vrf CUSTOMER_A 0.0.0.0 0.0.0.0 192.168.1.254
show ip route vrf CUSTOMER_A
ping vrf CUSTOMER_A 192.168.1.254
```

Modern IOS-XE syntax uses `vrf definition` + `address-family ipv4` instead of the legacy `ip vrf`.

21.4 MPLS & VXLAN (conceptual overview)

- **MPLS**: labels packets at ingress (LER) for fast label-switched forwarding through the core (LSR) without a full IP lookup at every hop; underpins L3VPN services.
- **VXLAN**: encapsulates L2 frames in UDP (port 4789) to extend broadcast domains across L3 (data-center overlay, used heavily with EVPN control-plane in modern fabrics like Cisco ACI).

Chapter 22

Automation: Netmiko, EEM, and Modern IOS-XE

22.1 Embedded Event Manager (EEM)

Reactive automation directly on the device.

```
event manager applet INTERFACE-DOWN
  event syslog pattern "LINEPROTO-5-UPDOWN.*GigabitEthernet0/1.*down"
  action 1.0 syslog msg "ALERT: Gi0/1 went down!"
  action 2.0 cli command "enable"
  action 3.0 cli command "interface gi0/1"
  action 4.0 cli command "shutdown"
  action 5.0 cli command "no shutdown"
```

22.2 Netmiko (Python-based automation — industry standard)

```
from netmiko import ConnectHandler

device = {
    "device_type": "cisco_ios",
    "host": "192.168.1.1",
    "username": "admin",
    "password": "Cisco123",
    "secret": "Cisco123",
}

conn = ConnectHandler(**device)
conn.enable()
output = conn.send_command("show ip interface brief")
print(output)

config_commands = ["interface loopback99", "ip address 9.9.9.9 255.255.255.255"]
conn.send_config_set(config_commands)
conn.save_config()
conn.disconnect()
```

22.3 RESTCONF / NETCONF (IOS-XE, model-driven programmability)

```
show restconf
restconf
netconf-yang
```

Modern network automation increasingly favors **YANG models** over screen-scraping CLI output — worth learning Ansible's `ios_*` modules and `ncclient/pyang` once comfortable with the CLI basics above.

Chapter 23

Appendix: Cheat Sheets

23.1 Subnetting Quick Reference

CIDR	Mask	Hosts	Block Size
/24	255.255.255.0	254	256
/25	255.255.255.128	126	128
/26	255.255.255.192	62	64
/27	255.255.255.224	30	32
/28	255.255.255.240	14	16
/29	255.255.255.248	6	8
/30	255.255.255.252	2	4
/31	255.255.255.254	2 (RFC 3021, P2P)	2
/32	255.255.255.255	1 (host route)	1

23.2 Common Well-Known Ports

Port	Protocol	Service
20/21	TCP	FTP (data/control)
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
67/68	UDP	DHCP (server/client)
69	UDP	TFTP
80	TCP	HTTP
110	TCP	POP3
123	UDP	NTP
143	TCP	IMAP
161/162	UDP	SNMP / SNMP Trap
179	TCP	BGP
443	TCP	HTTPS
445	TCP	SMB
514	UDP	Syslog

Port	Protocol	Service
3389	TCP	RDP

23.3 Command Reference by Task

Task	Command
Enter privileged mode	<code>enable</code>
Enter global config	<code>configure terminal</code>
Set hostname	<code>hostname <name></code>
Assign IP to interface	<code>ip address <ip> <mask></code>
Bring interface up	<code>no shutdown</code>
Save config	<code>copy running-config startup-config / wr</code>
View routing table	<code>show ip route</code>
View interfaces summary	<code>show ip interface brief</code>
Create VLAN	<code>vlan <id> / name <name></code>
Assign access port	<code>switchport mode access / switchport access vlan <id></code>
Configure trunk	<code>switchport mode trunk</code>
Static route	<code>ip route <net> <mask> <next-hop></code>
Enable OSPF	<code>router ospf <id> / network <net> <wildcard> area <area></code>
Enable EIGRP	<code>router eigrp <as> / network <net> <wildcard></code>
Standard ACL	<code>access-list <1-99> permit/deny <src> <wildcard></code>
Extended ACL	<code>access-list <100-199> permit/deny <proto> <src> <wc> <dst> <wc> eq <port></code>
PAT/NAT overload	<code>ip nat inside source list <acl> interface <int> overload</code>
DHCP pool	<code>ip dhcp pool <name> / network / default-router</code>
HSRP VIP	<code>standby <grp> ip <vip></code>
Port security	<code>switchport port-security</code>
SSH setup	<code>crypto key generate rsa + ip ssh version 2</code>

23.4 Suggested Mastery Path

1. Master CLI navigation, basic device config, interfaces, and subnetting cold — this is non-negotiable foundation.

2. Build small GNS3/EVE-NG/Packet Tracer labs for VLANs + trunking + inter-VLAN routing until muscle memory forms.
3. Layer in STP, EtherChannel, and FHRP on redundant topologies.
4. Get comfortable with static routing, then OSPF single-area, then multi-area, then EIGRP, then BGP basics.
5. Add ACLs, NAT, and DHCP to a working topology — simulate real branch/campus scenarios.
6. Layer 2 & device security (port-security, DHCP snooping, AAA, SSH hardening).
7. WAN technologies + VPNs, then IPv6 parallel-track everything you did in IPv4.
8. Finish with QoS, IOS file/image management, password recovery drills, and automation (Netmiko/EEM) to round out real-world operational skill.

Practice discipline: for every topic above, don't just read the commands — build the topology, break it on purpose, and use the **show/debug** commands to diagnose and fix it. That debugging loop is what actually builds CCNA/CCNP-level mastery.